(L)297mm

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pharmaniaga®

Dobutamine 12.5mg/mL Concentrate for Solution for Infusion

COMPOSITION
Each 20ml contains Dobutamine Hydrochloride 280.20mg is equivalent to 250.00 mg, Sodium metabisulphate 4.40 mg, Hydrochloride acid q.s., Sodium Hydroxide q.s. and Water for Injection q.s. to 20mL.

DESCRIPTION
Before and after dilution: A colourless to pale yellow solution.

PHARMACODYNAMICS
Adult population
Dobutamine directly stimulates β-adrenergic receptors and is generally considered a selective β1-adrenergic agonist, but the mechanisms of action of the drug are complex. It is believed that the β-adrenergic effects result from stimulation of adenyl cyclase activity. In therapeutic doses, dobutamine also has mild β2 - and α1 - adrenergic receptor agonist effects, which are relatively balanced and result in minimal net direct effect on systemic vasculature. Unlike dopamine, dobutamine does not cause release of endogenous norepinephrine. The main effect of therapeutic doses of dobutamine is cardiac stimulation. While the positive inotropic effect of the drug on the myocardium appears to be mediated principally via β1-adrenergic stimulation, experimental evidence suggests that α1-adrenergic stimulation may also be involved and that the α1-adrenergic activity results mainly from the (-) -stereoisomer of the drug.

The β1-adrenergic effects of dobutamine exert a positive inotropic effect on the myocardium and result in an increase in cardiac output due to increased myocardial contractility and stroke volume in healthy individuals and in patients with congestive heart failure. Increased left ventricular filling pressure decreases in patients with congestive heart failure. In therapeutic doses, dobutamine causes a decrease in peripheral resistance; however, systolic blood pressure and pulse pressure may remain unchanged or be increased because of augmented cardiac output. With usual doses, heart rate is usually not substantially changed. Coronary blood flow and myocardial oxygen consumption are usually increased because of increased myocardial contractility.

Electrophysiologic studies have shown that dobutamine facilitates atrio-ventricular conduction and shortens or causes no important change in intraventricular conduction. The tendency of dobutamine to induce cardiac arrhythmias may be slightly less than that of dopamine and is considerably less than that of isoproterenol or other catecholamines. Pulmonary vascular resistance may decrease if it is elevated initially and mean pulmonary artery pressure may decrease or remain unchanged. Unlike dopamine, dobutamine does not seem to affect dopaminergic receptors and causes no renal or mesenteric vasodilatation; however, urine flow may increase because of increased cardiac output.

PHARMACOKINETICS
Adult population
Absorption: Orally administered dobutamine is rapidly metabolised in the GI tract. Following IV administration, the onset of action of dobutamine occurs within 2 minutes. Peak plasma concentrations of the drug and peak effects occur within 10 minutes after initiation of an IV infusion. The effects of the drug cease shortly after discontinuing an infusion.

Distribution: It is not known if dobutamine crosses the placenta or is distributed into milk.

Elimination: The plasma half-life of dobutamine is about 2 minutes. Dobutamine is metabolised in the liver and other tissues by catechol-o-methyltransferase to an inactive compound, 3-0-methydobutamine and by conjugation with glucuronic acid. Conjugates of dobutamine and 3-0-methyldobutamine are excreted mainly in urine and to a minor extent in faeces.

INDICATION
Adult population
Dobutamine 12.5mg/mL Concentrate for Solution for Infusion is indicated in adults who require short-term treatment of cardiac failure secondary to acute myocardial infarction or cardiac surgery.

RECOMMENDED DOSAGE
Administration
Because of its short half-life, Dobutamine must be administered as a continuous intravenous infusion. This is most reliably accomplished using a precision volume control intravenous infusion set.

Following the initiation of a constant rate infusion or upon changing the rate, a steady-state Dobutamine plasma concentration is achieved within approximately 10 minutes. Thus, loading doses or bolus injections are not necessary are not recommended.

Dosage unit
Most reports on Dobutamine hydrochloride have expressed the dose in relation to body mass, for example, mcg/kg/min. This practice is useful to relate doses in infants and children to those in adults. Among adults, body mass has little influence on the effect of Dobutamine hydrochloride; since the dosage should be titrated in each patient, adults may be easily dosed with mcg/min units. The dosage of Dobutamine hydrochloride may be initiated at 100-200 mcg/min and increased gradually to 1000-2000 mcg/min or greater, depending on the clinical and hemodynamic response of the individual patient.

Dobutamine hydrochloride injection must be diluted with a compatible IV solution before administration; 20 mL of Dobutamine Hydrochloride 12.5mg/mL injection concentrate should be diluted in at least 50 mL of diluent.

Recommended Dosage
The rate of infusion needed to increase cardiac output usually ranges from 2.5 to 10 mcg/kg/min (see table). Frequently, doses up to 20 mcg/kg/min are required for adequate hemodynamic improvement. On rare occasions, infusion rates up to 40 mcg/kg/min have been reported.

Rate of Infusion for Concentrations of 250, 500 and 1,000 mg/L

Drug Delivery Rate mcg/kg/min	Infusion Delivery Rate		
	250 mg/L * (mL/kg/min)	500 mg/L** (mL/kg/min)	1,000 mg/L*** (mL/kg/min)
2.5	0.01	0.005	0.0025
5.0	0.02	0.01	0.005
7.5	0.03	0.015	0.0075
10.0	0.04	0.02	0.01
12.5	0.05	0.025	0.0125
15.0	0.06	0.03	0.015

*250 mg/L of admixture (add one 20mL vial of 12.5 mg/mL of Dobutamine Injection USP to obtain a final admixture volume of 1 L).

**500 mg/L or 250 mg/500 mL of admixture (add one 20 mL vial of 12.5 mg/mL of Dobutamine Injection USP to obtain a final admixture volume of 500 mL).

***1,000 mg/L or 250 mg/250 mL of admixture (add one 20mL vial of 12.5 mg/mL of Dobutamine Injection USP to obtain a final admixture volume of 250 mL).

The rate of administration and the duration of therapy should be adjusted according to the patient's response, as determined by the following clinical indicators: hemodynamic parameters, such as heart rate and rhythm, arterial pressure, and whenever possible, cardiac output and measurements of ventricular filing pressure (central venous, pulmonary capillary wedge and left atrial), and signs of pulmonary congestion or organ perfusion (urine flow, skin temperature and mental status).

Concentrations up to 5000 mg/L have been administered to humans (250 mg/50 ml). The final volume administered should be determined by the fluid requirements of the patient.

Rather than abruptly discontinuing therapy with Dobutamine hydrochloride, it is often advisable to decrease the dosage gradually.

Compatibilities
Pharmaniaga Dobutamine 12.5 mg/mL Concentrate for Solution for Infusion is compatible with the following infusion solutions:

- 0.9% Sodium Chloride
- 5% Dextrose
- 10% Dextrose
- 0.9% Sodium Chloride and Dextrose 5% (NSD5)
- Lactated Ringer's

The solution will remain stable for 24 hours at room temperature (30°C) after diluted with the compatible solutions.

Solution of Dobutamine hydrochloride may have a pink discoloration. This discoloration, which will increase with time, results from a slight oxidation of the drug. However, there is no significant loss of drug potency within the recommended storage times for solutions of the drugs.

ROUTE OF ADMINISTRATION
For intravenous (IV) infusion use.

CONTRAINDICATIONS
Dobutamine hydrochloride is contraindicated in patients with idiopathic hypertrophic subaortic stenosis or with known hypersensitivity to Dobutamine hydrochloride or any of the excipients.

WARNINGS AND PRECAUTIONS
Increase in Heart Rate or Arterial Blood Pressure
Dobutamine hydrochloride may cause an increase in heart rate or blood pressure, especially in systolic pressure. Approximately 10 percent of patients in clinical studies have had rate increases of 30 beats/minute or more, and about 7.5 percent have a 50 mm Hg or greater increase in systolic pressure. Reduction of dosage usually reverses these effects promptly. Patients with pre-existing hypertension are more likely to develop and exaggerated pressor response.

Increased Atrioventricular Conduction
Because Dobutamine hydrochloride facilitates atrioventricular conduction, patients with atrial flutter or fibrillation may develop rapid ventricular responses.

Ectopic Activity
Dobutamine hydrochloride may precipitate or exacerbate ventricular ectopic activity; rarely has it caused ventricular tachycardia or fibrillation.

Impaired Ventricular Filling and Ventricular Outflow Obstruction
No improvement may be observed in the presence of marked mechanical obstruction, such as severe valvular aortic stenosis.

Inotropic agents, including Dobutamine hydrochloride, do not improve haemodynamic in most patients with mechanical obstruction that hinders either ventricular filling or outflow, or both. Inotropic response may be inadequate in patients with markedly reduced ventricular compliance. Such conditions are present in cardiac tamponade, valvular aortic stenosis, and idiopathic hypertrophic subaortic stenosis. Beneficial inotropic effects may be seen in some patients if the heart is dilated or under excessive effect of ß- adrenergic receptor antagonists.

Hypersensitivity
Reactions suggestive of hypersensitivity associated with the administration of Dobutamine hydrochloride, including skin rash, fever, eosinophilia, and bronchospasm, have been reported occasionally.

Paediatric Population
The safety and efficacy of Dobutamine Hydrochloride for use in children have not been studied.

Dobutamine has been administered to children with low- output hypoperfusion states resulting from decompensated heart failure, cardiac surgery and cardiogenic and septic shock. Some of the haemodynamic effects of dobutamine hydrochloride may be quantitatively or qualitatively different in children as compared to adults. Increment in heart rate and blood pressure appear to be more frequent and intense in children. Pulmonary wedge pressure may not decrease in children, as it does in adults, or it may actually increase, especially in infants less than one year old. The neonate cardiovascular system has been reported to be less sensitive to dobutamine and hypotensive effect seems to be more often observed in adult patients than in small children.

Accordingly, the use of dobutamine in children should be monitored closely, bearing in mind these pharmacodynamic characteristics.

General
In patients who have atrial fibrillation with rapid ventricular response, a digitalis preparation should be used prior to instituting therapy with Dobutamine hydrochloride. During the administration of Dobutamine hydrochloride, as with any parenteral catecholamine, heart rate and rhythm, arterial blood pressure, and infusion rate should be monitored closely. When initiating therapy, electrocardiographic monitoring is advisable until a stable response is achieved.

Hypovolaemia should be corrected before treatment with Dobutamine hydrochloride is instituted.

During the administration of Dobutamine Hydrochloride, as with any adrenergic agent, ECG and blood pressure should be continuously monitored. In addition, pulmonary artery wedge pressure and cardiac output should be monitored whenever possible to aid in the safe and effective infusion of Dobutamine Hydrochloride.

Animals studies indicate that Dobutamine Hydrochloride may be ineffective if the patients has recently received a β-blocking drug. In such a case, the peripheral vascular resistance may increase.

Usage for Heart Failure Complicating an Acute Myocardial Infarction
Although the treatment of heart failure and the reduction in cardiac diameter will decrease myocardial oxygen consumption, there is still concern that the use of any inotropic agent may

attn	customer **Pharmaniaga LifeScience Sdn Bhd**	date **19.03.2021**
1 colours	Black	Please Chop & Sign For Approval
size (L)**297 x** (W)**210mm**		
description **D3-PIL Dobutamine insert – Front**		
FocusPrint SDN BHD	t: 603-8766 6030 f: 603-8766 6033 (Factory) website: www.focusprint.com.my	
artwork prepared by: cynthia yap	email: graphic@focusprint.info (Graphic Dept)	**date:**

ARTWORK LOG

Version no.	Date	Reason for Change
01	06.08.2019	- Amend PM number.
02	26.08.2019	- Remove PM.
03	08.09.2020	- Amend wording and revision date.
04	11.09.2020	- Amend wording.
05	17.09.2020	- Amend wording and revision date.
06	19.03.2021	- Add in Protect from light.

IMPORTANT ! **Please note that this colour print is for visual purpose only. Output colour might differ in actual production.**

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increase myocardial oxygen demand and the size of an infarction by intensifying ischemia. There is concern that any agent which increases contractile force and heart rate may increase the size of an infarction by intensifying ischaemia, but it is not known whether dobutamine does so in man. However, animal studies have shown that massive doses of 30 mg/kg/min infused for 72 hours have produced irreversible myocardial damage.

Dobutamine hydrochloride does not have an adverse effect on the myocardium when used in doses that do not cause excessive increments in heart rate or arterial pressure. The dose of Dobutamine hydrochloride should be titrated to prevent an excessive increase in heart rate and systolic blood pressure.

Usage in Hypotension
When hypotension is largely attributable to decreased cardiac output and coincides with an elevated ventricular filling pressure, the infusion of Dobutamine hydrochloride may help restore the pressure.

As volume is replenished in the treatment of acute hypoperfusion states and there is an increase in pulmonary wedge or central venous pressures but with no increase in cardiac output and arterial pressure, Dobutamine hydrochloride may improve output and help restore the arterial pressure.

In general, when mean arterial blood pressure is less than 70-mm Hg in the absence of an increased ventricular filling pressure, hypovolaemia may be present and would require treatment with appropriate volume repleting solutions before Dobutamine hydrochloride is given. Hypovolaemia should be corrected with suitable volume expanders before treatment with Dobutamine Hydrochloride is instituted.

If arterial blood pressure remains low or decreases progressively during administration of Dobutamine hydrochloride despite adequate ventricular filling pressure and cardiac output, consideration may be given to the concomitant use of a peripheral vasoconstrictor agent such as dopamine or norepinephrine.

Anaesthetics
The myocardium may be sensitized to the effect of dobutamine by cyclopropane or halogenated hydrocarbon anesthetics, and these should be avoided.

Reactions at Sites of Intravenous Infusion
Phlebitis has occasionally been reported. Local inflammatory changes have been described following inadvertent infiltration. Isolated cases of cutaneous necrosis have been reported.

Warning for Sodium Metabisulphite
This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites

INTERACTIONS WITH OTHER MEDICAMENTS
Halogenated anaesthetics:
Although it is less likely than adrenaline to cause ventricular arrhythmias, Pharmaniaga Dobutamine 12.5 mg/ml concentrate for solution for infusion should be used with great caution during anaesthesia with cycloproprane, halothane and other halogenated anaesthetics.

Entacapone:
The effects of Pharmaniaga Dobutamine 12.5 mg/ml concentrate for solution for infusion may be enhanced by entacapone.

Beta-blockers:
The inotropic effect of dobutamine stems from stimulation of cardiac beta-1 receptors, this effect is reversed by concomitant administration of beta-blockers. Dobutamine has been shown to counteract the effect of beta-blocking drugs. In therapeutic doses, dobutamine has mild alpha1-and beta2-agonist properties. Concurrent administration of a non-selective beta-blocker such as propranolol can result in elevated blood pressure, due to alpha-mediated vasoconstriction, and reflex bradycardia. Beta-blockers that also have alpha-blocking effects, such as carvedilol, may cause hypotension during concomitant use of dobutamine due to vasodilation caused by beta-2 predominance.

Incompatibilities
Because of potential physical incompatibilities, it is recommended that Dobutamine hydrochloride not be mixed with other drugs in the same solution. Dobutamine hydrochloride are incompatible with Sodium bicarbonate injection, or any other strongly alkaline solutions and should not be used in conjunction with other drugs or diluents containing both Sodium bisulphites and ethanol.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION
Reproduction studies in rats and rabbits have revealed no evidence of impaired fertility, harm to the foetus, or teratogenic effects due to dobutamine. As there are no adequate and well-controlled studies in pregnant women, and as animal reproduction studies are not always predictive of human response, dobutamine should not be used during pregnancy unless the potential benefits outweigh the potential risks to the foetus.

ADVERSE EFFECTS

Immune system disorders

Not known	:	Hypersensitivity reactions including rash, fever, eosinophilia and bronchospasm have been reported. Anaphylactic reactions and severe life-threatening asthmatic episodes may be due to sulphite sensitivity.

Blood and lymphatic system disorders

Common	:	Eosinophilia, inhibition of thrombocyte aggregation (only when continuing infusion over a number of days).

Metabolism and nutrition disorders

Very rare	:	Hypokalaemia

Nervous system disorders

Common	:	Headache
Not known	:	Paraesthesia, tremor, myoclonic spasm. Myoclonus has been reported in patients with severe renal failure receiving dobutamine

Cardiac disorders/ Vascular disorders

Very common	:	Increase of the heart rate by ≥ 30 beats/min
Common	:	Blood pressure increase of ≥ 50 mmHg. Patients suffering from arterial hypertension are more likely to have a higher blood pressure increase. Blood pressure decrease, ventricular dysrhythmia, dose-dependent ventricular extrasystoles. Increased ventricular frequency in patients with atrial fibrillation. These patients should be digitalized prior to dobutamine infusion. Vasoconstriction in particular in patients who have previously been treated with beta receptor blockers. Anginal pain, palpitations
Uncommon	:	Ventricular tachycardia, ventricular fibrillation
Very rare	:	Bradycardia, myocardial ischaemia, myocardial infarction, cardiac arrest
Not known	:	Electrocardiogram ST segment elevation Decrease in pulmonary capillary pressure Eosinophilic myocarditis has been noted in explanted hearts of patients who had undergone treatment with multiple medications including dobutamine or other inotropic agents prior to transplantation. Children: pronounced increase of heart rate and/or blood pressure as well as a lower decrease of the pulmonary capillary pressure than adults. Increase of pulmonary capillary pressure in children under 1.

Gastrointestinal disorders

Not known	:	Nausea

Psychiatric disorders

Not known	:	Restlessness, feeling of heat and anxiety

Renal and urinary disorders

Not known	:	Urinary urgency

Paediatric population

The undesirable effects include elevation of systolic blood pressure, systemic hypertension or hypotension, tachycardia, headache, and elevation of pulmonary wedge pressure leading to pulmonary congestion and edema, and symptomatic complaints.

Long term safety

Infusions for up to 72 hours have revealed no adverse effects other than those seen with shorter infusions.

There is evidence that partial tolerance develops with continuous infusions of Dobutamine 12.5 mg/ml concentrate for solution for infusion for 72 hours or more; therefore, higher doses may be required to maintain the same effects.

Dobutamine stress echocardiography

Cardiac disorders/ Vascular disorders

Very common	:	Pectoral anginal discomfort, ventricular extra-systoles with a frequency of > 6/min
Common	:	Supraventricular extrasystoles, ventricular tachycardia
Uncommon	:	Ventricular fibrillation, myocardial infarction
Very rare	:	Occurrence of a second degree atrioventricular block, coronary vasospasms.
	:	Hypertensive/hypotensive blood pressure decompensation, occurrence of an intracavitary pressure gradient, palpitations.
Not known	:	Stress cardiomyopathy, Left ventricular outflow tract obstruction, Fatal cardiac rupture

Respiratory system, thoracic and mediastinal disorders

Common	:	Bronchospasm, shortness of breath

Gastrointestinal disorders

Common	:	Nausea

Skin and subcutaneous tissue disorders

Common	:	Exanthema
Very rare	:	Petechial bleeding

Musculoskeletal and connective tissue disorders

Common	:	Chest pain

Renal and urinary disorders

Common	:	Increased urgency at high dosages of infusion

General disorders and administration site conditions

Common	:	Fever, phlebitis at the injection site In case of accidental paravenous infiltration, local inflammation may develop
Very rare	:	Cutaneous necrosis

OVERDOSE AND TREATMENT
Overdoses of Dobutamine have been reported rarely. The following is provided to serve as a guide if such an overdose is encountered.

Signs and Symptoms
Toxicity from Dobutamine hydrochloride is usually due to excessive cardiac ß-receptor stimulation. The duration of action of Dobutamine hydrochloride is generally short (T½=2 minutes) because it is rapidly metabolized by catechol-O-methyltransferase. The symptoms of toxicity may include anorexia, nausea, vomiting, tremor, anxiety, palpitations, headache, shortness of breath, and angina and non-specific chest pain.

The positive inotropic and chronotropic effects of Dobutamine on the myocardium may cause hypertension, tachyarrhythmias, myocardial ischemia, and ventricular fibrillation. Hypotension may result from vasodilation. Unpredictable absorption may occur from the mouth and the gastrointestinal tract if dobutamine hydrochloride been ingested.

Treatment
In managing overdosage, consider the possibility of multiple drug overdoses, interaction among medicines, and unusual medicine kinetics in your patient.

The initial actions to be taken in a Dobutamine hydrochloride overdose are discontinuing administration, establishing an airway, and ensuring oxygenation and ventilation. Resuscitative measures should be initiated promptly. Severe ventricular tachyarrhythmias may be successfully treated with Propranolol or Lidocaine. Hypertension usually responds to a reduction in dose or discontinuation of therapy.

Protect the patient's airway and support ventilation and perfusion. If needed, meticulously monitor and maintain, within acceptable limits, the patient's vital signs, blood gases, serum electrolytes, etc.

Absorption of medicines from the gastrointestinal tract may be decreased by giving activated charcoal, which, in many cases, is more effective than emesis or lavage; consider charcoal instead of or in addition to gastric emptying. Repeated doses of charcoal over time may hasten elimination of some medicines that have been absorbed. Safeguard the patient's airway when employing gastric emptying or charcoal.

Forced diuresis, peritoneal dialysis, haemodialysis, or charcoal hemoperfusion have not been established as beneficial for an overdose of Dobutamine hydrochloride.

Effects on ability to drive and use machines
Not applicable in view of the indications for use and the short half-life of the drug.

INSTRUCTION FOR USE
Must be diluted prior used & for single use only.

STORAGE CONDITIONS
Store below 30°C.
Protect from light.

SHELF LIFE
The injections can be used within 24 months from the date of manufacture if kept as recommended.

Once opened discard any unused portion.

Pharmaniaga Dobutamine 12.5mg/mL Concentrate for Solution for Infusion is compatible with the following infusion solutions:

- 0.9% Sodium Chloride
- 5% Dextrose
- 10% Dextrose
- 0.9% Sodium Chloride and Dextrose 5% (NSD5)
- Lactated Ringer's

The solution will remain stable for 24 hours at room temperature (30°C) after diluted with the compatible solutions.

DOSAGE FORMS AND PACKAGING AVAILABLE
5x20mL vial (clear glass vial) of Pharmaniaga Dobutamine 12.5mg/ml Concentrate for Solution for Injection.

5 vials per unit PVC tray were packed in paper carton.

PRODUCT REGISTRATION HOLDER/MANUFACTURER
PHARMANIAGA LIFESCIENCE SDN BHD (198201002939)
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47100 Puchong, Selangor Darul Ehsan, Malaysia.

2002188

Revision Date: 27-August-2021

attn | customer **Pharmaniaga LifeScience Sdn Bhd** | date **19.03.2021**

1 colours Black

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size (L)**297** x (W)**210mm**
description **D3-PIL Dobutamine insert – Back**

FocusPrint SDN BHD t: 603-8766 6030 f: 603-8766 6033 (Factory)
website: www.focusprint.com.my

artwork prepared by: cynthia yap email: graphic@focusprint.info (Graphic Dept)

date:

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ARTWORK LOG

Version no.	Date	Reason for Change
01	06.08.2019	- Amend PM number.
02	26.08.2019	- Remove PM.
03	08.09.2020	- Amend wording and revision date.
04	11.09.2020	- Amend wording.
05	17.09.2020	- Amend wording and revision date.
06	19.03.2021	- Add in Protect from light.